



JUDGING STANDARDS

Reporting against the Achievement Standard

The annotated work samples in *Judging Standards* support teachers when reporting against the achievement standards; when giving assessment feedback; and when explaining the differences between one student's achievement and another's.

Grey highlighting identifies those aspects of the achievement standard addressed in the work sample. Annotations in black text refer to the assessment pointers while those in coloured text highlight additional, specific qualities evident in the work.

WORK SAMPLE: YEAR 10 SCIENCE

PERFORMANCE ASSOCIATED WITH GRADE B, REPRESENTING HIGH ACHIEVEMENT

SUMMARY OF TASK:

Gene technology: Stem cell research

Students selected an area of gene technology to research and presented their information on a poster using text and graphics. Part of the task involved identifying positives and negatives related to this type of technology.

YEAR 10 SCIENCE ACHIEVEMENT STANDARD

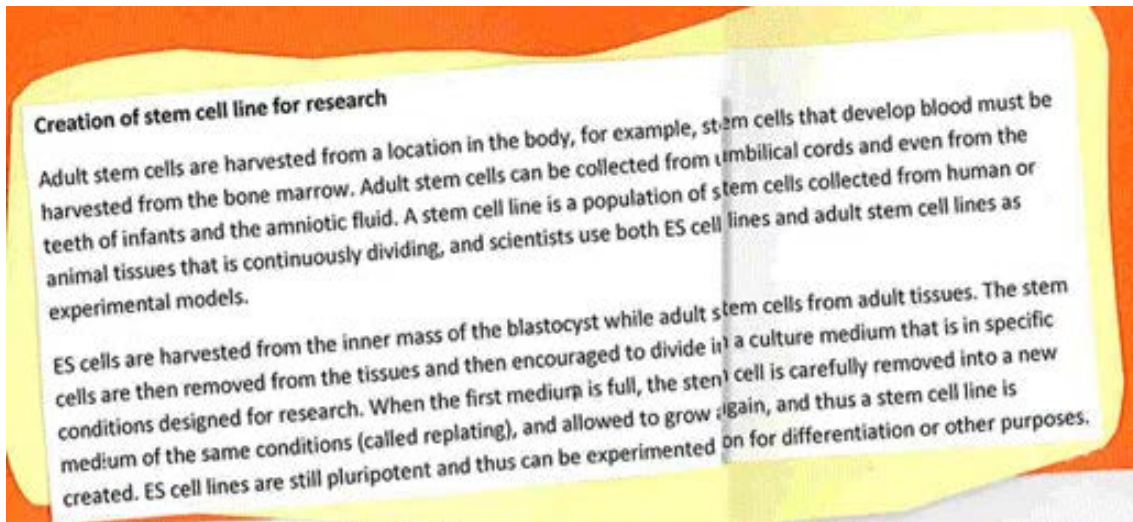
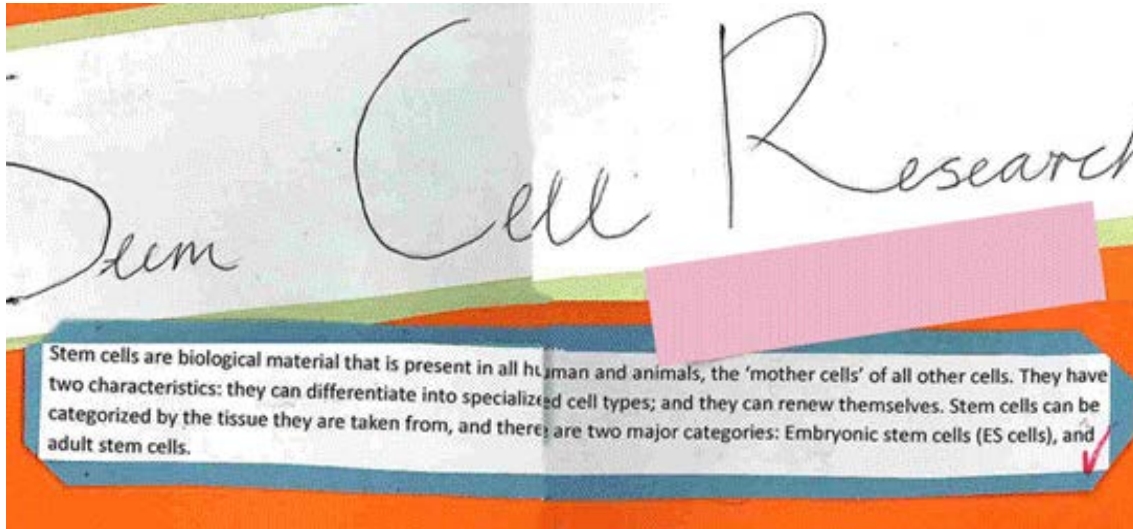
By the end of Year 10, students analyse how the periodic table organises elements and use it to make predictions about the properties of elements. They explain how chemical reactions are used to produce particular products and how different factors influence the rate of reactions. They explain the concept of energy conservation and represent energy transfer and transformation within systems. They apply relationships between force, mass and acceleration to predict changes in the motion of objects. Students describe and analyse interactions and cycles within and between Earth's spheres. They evaluate the evidence for scientific theories that explain the origin of the universe and the diversity of life on Earth. They explain the processes that underpin heredity and evolution. Students analyse how the models and theories they use have developed over time and discuss the factors that prompted their review.

Students develop questions and hypotheses and independently design and improve appropriate methods of investigation, including field work and laboratory experimentation. They explain how they have considered reliability, safety, fairness and ethical actions in their methods and identify where digital technologies can be used to enhance the quality of data. When analysing data, selecting evidence and developing and justifying conclusions, they identify alternative explanations for findings and explain any sources of uncertainty. Students evaluate the validity and reliability of claims made in secondary sources with reference to currently held scientific views, the quality of the methodology and the evidence cited. They construct evidence-based arguments and select appropriate representations and text types to communicate science ideas for specific purposes.

KEY WORDS:

Genes, DNA, Genetic manipulation, Genetic engineering

Student achievement is reported at the end of the semester or year using the letter grades and achievement descriptors. Letter grades and achievement descriptors should not be used to assess individual pieces of work.



Outlines scientific concepts and processes, using correct scientific terminology and knowledge, e.g. explains where stem cells are harvested from. Uses terms such as 'pluripotent' and 'amniotic fluid'.

Adult Stem Cells
Adult stem cells have far more practical use for the application than ES cells, and are utilized mostly for replacement and repair of damaged tissues and organs. They are multipotent – capable of giving rise to multiple, but limited, cells, and limited in their lineage differentiation. It was previously thought that adult stem cells are cells cannot differentiate into cells of a different lineage, but recently some adult stem cells have been discovered to be able to do just that; blood stem cells at all, but recently some adult stem cells have been able to give rise to both skeletal muscles and neurons.

Unlike ES cells, adult stem cells are much rarer and have been discovered in the following adult organs and tissues: bone marrow, peripheral blood, cornea, retina, skeletal muscle, brain, blood vessels, and used stem pancreas, epithelia of the skin and digestive system, the liver. The most commonly studied and used stem cells are hematopoietic stem cells from the bone marrow.

A blastocyst is an early embryo in the shape of a sphere. It has two layers – an outer layer that develops into the placental and other fetal tissue, and an inner cell mass.

Embryonic Stem cells
ES cells are taken from early stage embryos – the blastocyst. Embryos reach the blastocyst stage at around 4-5 days after fertilization, and they can contain up to 50 – 150 cells. Embryonic stem cells are pluripotent, meaning that they can give rise to all types of human cells except the placenta and fetal support tissues.

To acquire ES cells, a human embryo must be acquired. The cells are taken from the inner cell mass of the blastocyst which requires its isolation, and this raises ethical issues as the embryo is destroyed in the process. Most of the embryos used are from surplus embryos left from assisted reproductive technology (ART), though they can be created specifically through somatic cell nuclear transfer, also known as cloning.

Currently, ES cells have only been used in research though its future holds great potential. The application of the cells is still experimental and no current use for it has been found.

Stem Cell Research
Stem cell research focuses on:
- the investigation of molecular characteristics of all stem cell types;
- improving methods for culturing; and most importantly
- the development of one cell into many others of different varieties.

In ES cells, the RNA has not yet begun triggering proteins into cell development and therefore remains unspecialized. If we can control the specialization of ES cells, then ideally we will be able to grow any type of human tissue that we need. However, uses for embryonic stem cells are still mainly done with adult stem cells, even though they are much more, limited.

VS.

DNA		RNA	
Deoxyribose sugar	Double helix structure	Ribose sugar	Single Strand
Mainly used for long term storage and transmission of genetic information		Mainly used to transfer genetic codes needed for creation of proteins	

RNA
Adenine
Uracil
Cytosine
Guanine
Ribose
Phosphate

Cell	Potency	Cell Potency
Zygote	Totipotent	-can give rise to all types of cells
Embryonic Stem cells	Pluripotent	-can give rise to all types of cell except the placenta and foetal support tissues
Adult Stem Cell	Multipotent	-can give rise to multiple, but limited number of cells
Progenitor	Unipotent	Able to differentiate into one type of cell
Differentiated cell	No further differentiation	

Draws diagrams to represent scientific models in support of explanations, e.g. RNA.

Outlines scientific concepts and processes, using correct scientific terminology and knowledge, e.g. zygote; blastocyst; progenitor.

Differentiation of Stem Cells

Though each cell have a subset of all the genes in the genome of its organism, only specific genes are active. Totipotent cells have nearly its entire genes active, while pluripotent cells have less, and multipotent cells even less. Cell differentiation is the transfer from one gene pattern to another, and controlling cell differentiation is the switching on of specific gene patterns.

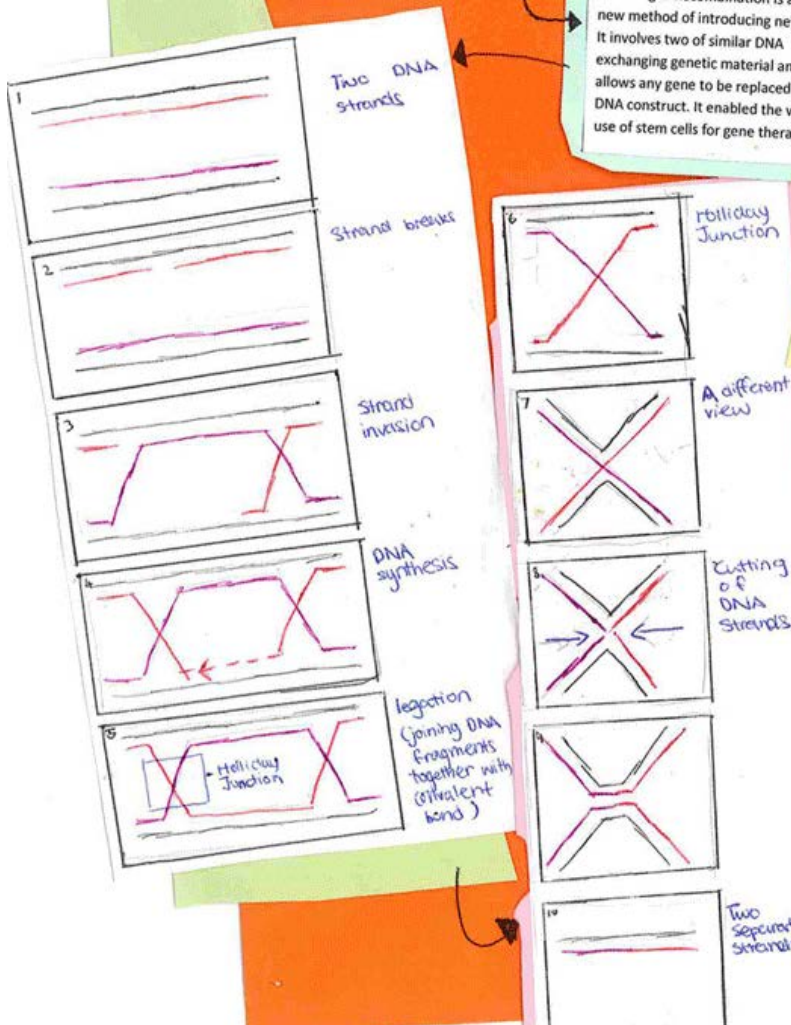
There are many ways to direct stem cells to differentiate into a specific cell, and many are still being developed. In order for cells to differentiate manually, researchers must imitate the signals found in the human body to stimulate the stem cells. This can be done by using drug molecules to push the cells into differentiating, and the introduction of DNA, RNA or proteins deliberately.

Homologous Recombination

Homologous Recombination is a fairly new method of introducing new DNA. It involves two of similar DNA exchanging genetic material and allows any gene to be replaced with a DNA construct. It enabled the wide use of stem cells for gene therapy.

Electroporation

Electroporation is one of the many ways of introducing molecules into mammalian cells. It uses a strong electrical pulse, which surpasses the membrane's capacitance (ability to store electrical charge), lasting from a few nanoseconds to a millisecond to overcome barriers of the cell membrane and causes the reversible breakdown of the membrane. This creates pores which can allow nucleic acids different molecules through to create the desired effect in stem cells.

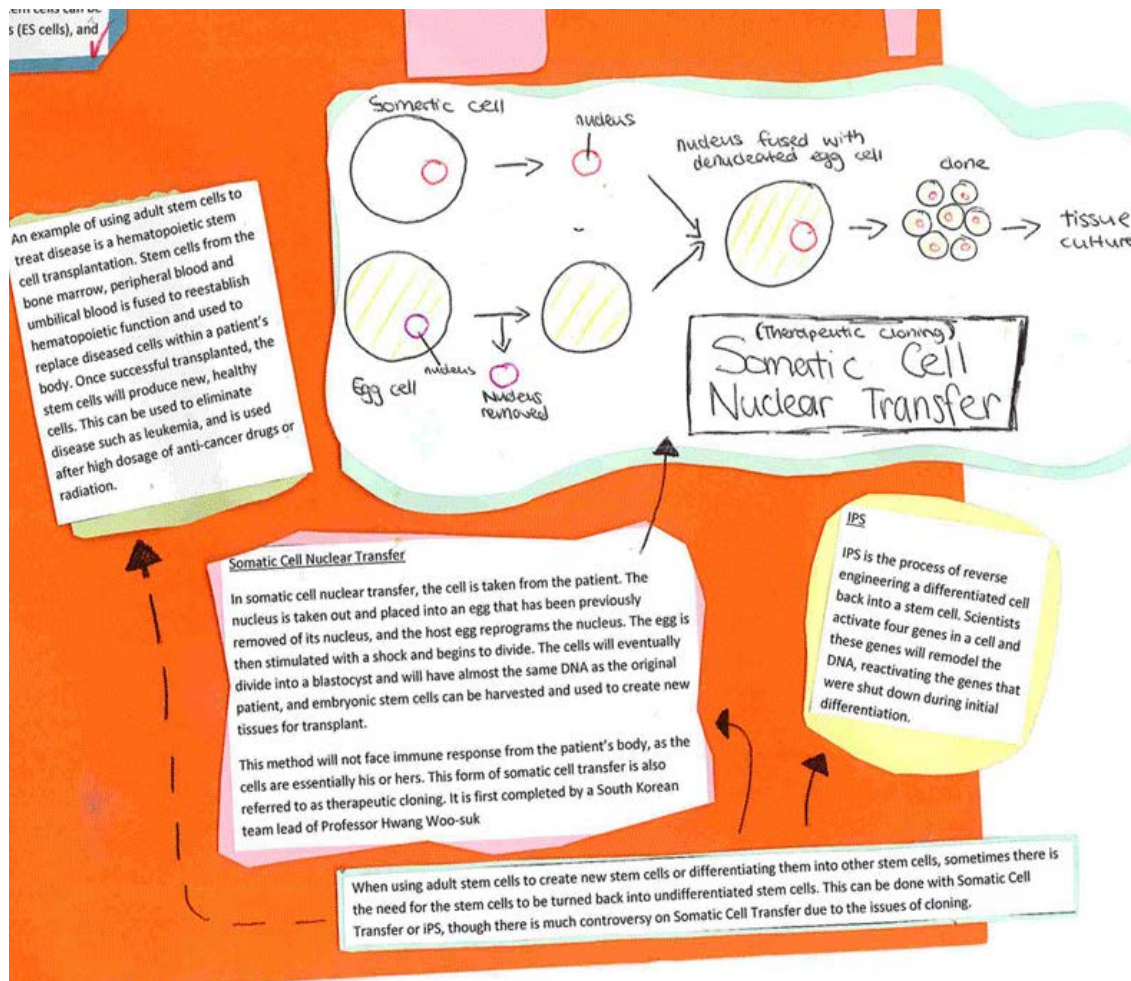


Names and explains a process, e.g. describes how DNA is altered.

Creation

Adult s
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ES cel
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Draws diagrams to represent scientific models in support of explanations, e.g. Somatic Cell Nuclear Transfer.

Implications of Stem Cell Research

Embryonic stem cells have the potential to grow into any type of cell in the body and could possibly be grown into organs and tissues for transplants of all kinds. Therapeutic cloning would allow scientists to grow organs and tissues in vitro (outside the body) and when it is transplanted into the patient, his or her body and immunity system will not reject it, as it is essentially his or her own tissues with the same DNA. Currently untreatable diseases could also be treated though the replacement of stem cells to produce healthy ones instead. ES cells could potentially heal billions of people, and cure illness including: diabetes, cancer, cardiovascular disease, Alzheimer's, Parkinson's, autoimmune disease, burn victims, blood diseases, leukemia, spinal injuries and even more.

Studies with ES cells could also help us to understand the entire human body and its structure, as embryonic stem cell is pluripotent and is essentially the 'mother' of most other cell types. Therapeutic cloning could also allow us to study diseases in vitro and experiment the effects of drugs and new treatments on live tissue without animal or human testing, essentially ending another controversial issue.

Using embryonic stem cells could also save future money on medical bills. In 2005, Medicare spent \$91 billion on beneficiaries with dementia related diseases, and that is just Medicare alone. Curing of the disease would save billions every year.

Negative implications could be science's slipping into dehumanizing practices. If embryonic stem cell research becomes acceptable, the next controversial issue could be human cloning, and maybe research on live humans. Science would quickly spiral out of control.

Considers implications of particular research and technologies, e.g. outlines positive and negative aspects of the stem cells research.

Pros	Cons
Embryonic stem cell research has the potential to treat almost all diseases (as mentioned above in the 'implications')	Science slipping into dehumanizing practices
Use of embryos from abortions and IVF that would've been thrown away other wise	ES cells could result in tumor and cancer cells if not cultured and maintained carefully
Saving millions of lives	Destruction of human life
Stem cells can continuous renew themselves and are in a sense unlimited	ES cells have not cured anything so far, and many of the implications could be false
No need for animal/human testing	Harvesting eggs from women can be dangerous. The procedure includes multiple injections of hormones and minor surgeries and can even result in Ovarian Hyper-stimulation Syndrome.
Further understanding of human race, including our origins, how our body functions and why it functions that way	Ovarian Hyper-stimulation syndrome could lead to rapid weight gain, accumulation of serous fluid in the spaces between tissues, even respiratory difficulties.
Economic implications could save a lot of money	Exploitation of women for eggs – the use of misleading or coercive language and other unethical methods to obtain eggs. Eg. The Korean scientist Hwang used over 2000 eggs to successfully create one Somatic Cell Transferred cell. It was discovered that he the women
adult stem cells are already used to save lives	Misuse of knowledge such as cloning for personal gain